

falsify — Phase 8

Time-series momentum, measured against Moskowitz-Ooi-Pedersen (2012), and the state of every specification document the build was given.

Commit 196a5ca · 2026-08-24 · 439/449 tests collected (10 deselected) in 15.29s

What Phase 8 was for

PLAYBOOK names time-series momentum as **“a free calibration check against the literature”**: published Sharpe near 0.8, and **“if yours comes out at 3.0 on SPY, you have a bug.”** The value of the check is that it is falsifiable before the code is written — the number to beat was fixed by someone else, years ago, and cannot be adjusted after the fact.

It is an upper bound, not a target, and the distinction matters. The published 0.8 is a diversified basket of 58 futures across four asset classes, each volatility-scaled and equally weighted. Most of that Sharpe is diversification. A single equity index cannot reproduce it, and a single equity index that appeared to would be reporting luck.

On real SPY, 2015-2024, zero cost

strategy	SR	±SE	turns/yr	HAC t
TSMomentum(12m,1m)	+0.606	0.340	1.56	+1.93
MOP form: VolTarget(0.40, cap 2)	+0.673	0.340	3.27	+2.10
TSMomentum(12m,3m)	+0.203	0.336	1.56	+0.65
TSMomentum(6m,1m)	+0.166	0.327	3.38	+0.55
BuyAndHold	+0.830	0.325	0.00	+2.80
MACrossover(20,50)	+0.217	0.319	9.41	+0.72

0.606 against a published 0.8 looks like a hit and is not one. The standard error is 0.340 — wide enough to contain 0.8, 0.606 and zero at once. What the check licenses is the negative: nothing is near 3.0, so nothing indicates a bug.

Buy-and-hold beat every configuration, at +0.830, and is the only strategy in the zoo whose Newey-West t-statistic clears 2. That is stated here rather than buried, and it is the same answer the A4 live test reached from the other direction.

The synthetic triple: null, power, adversary

16 paths of 1,600 bars each, TimeSeriesMomentum(12m, 1m), zero cost.

process	SR	±SE	SE from 0	verdict
GBM, mu = 0.08 (the null)	-0.141	0.145	-1.0	no edge, as it must be
GBM, mu = 0.00 (the null)	-0.134	0.099	-1.4	no edge
persistent drift, psi = 0.98	+0.804	0.203	+4.0	power
persistent drift, psi = 0.99	+1.659	0.262	+6.3	power
stationary AR(1)	-1.041	0.102	-10.2	loses to its adversary

The last row is the one worth having. A trend follower loses at ten standard errors on the exact stationary process **CausalZScore** earns on — same engine, same costs, same measurement. An edge is a property of a process, not of a strategy.

The generator that had to be replaced

A power test needs a process with a trend in it, and the obvious construction — autocorrelate the returns — does not work. This was measured rather than reasoned about, after the first version was already written:

process	12-month TSMOM SR	±SE
GBM, no autocorrelation	+0.092	0.092
AR(1) RETURNS, $\phi = 0.10$	+0.041	0.086
AR(1) RETURNS, $\phi = 0.30$	+0.025	0.109
persistent DRIFT, $\psi = 0.95$	+0.516	0.169
persistent DRIFT, $\psi = 0.99$	+1.866	0.225
persistent DRIFT, $\psi = 0.995$	+2.572	0.257

Nothing, at a daily ϕ of 0.30 that no real market approaches. The reason is arithmetic: an AR(1) has autocorrelation ϕ^k , and 0.30^{252} is zero. A signal that integrates a year of returns cannot read a structure that has decayed within a fortnight.

Momentum is not lag-1 autocorrelation.

Had the power test shipped on that generator it would have failed the strategy while the strategy was correct — the worst kind of test, one that is wrong in the direction of looking rigorous. The last row also explains why 3.0 is a sound bug threshold: even a process far more obliging than any market only reaches 2.57.

Three defects found by running it

what	how it surfaced
lookback must be <code>window + 1</code>	The engine slices <code>signals[start - lag:]</code> and reads from exactly <code>lookback</code> . Declaring the window handed it one NaN and the run was refused.
G2's fixture was too short	TSMOM joined ZOO so G1 and G2 certify it. G2's 220 bars left zero reported bars against a 253-bar lookback — the fixture was wrong, not the gate. It is 400 now.
Vol targeting is not a no-op	Synthetic constant-vol data said the 40% cap binds everywhere. On SPY it changes the weight at exactly 65 of 2,261 bars — exactly the 65 above 40% vol — from 2020-03-18 to 2020-06-18.

That third one is the argument for live tests in one line: an overlay exercised only on synthetic data looks like dead code, because synthetic data has no volatility clustering for it to respond to.

Checklists

Every gate and every invariant the build was given, against what exists in the tree at this commit.

Gate set — PLAYBOOK Part 2

item	state	note
G1 causality	GREEN	500 taus x 20 seeds, both cut modes, four zoo strategies
G2 twin engines	GREEN	max relative deviation 0.000e+00; fixture widened to 400 bars
G3 analytic recovery	GREEN	simple and log Sharpe; $\beta_{\text{simple}} - \beta_{\text{log}} = \sigma/2$
G4 zero-cost identity	GREEN	exact float equality
G5 cost monotonicity	GREEN	break-even cost finite and reported
G6 null calibration	GREEN	bounds set from a 15-world replication study
G7 leakage trap	GREEN	G1 flags the injected leak
G8 walk-forward	GREEN	purge and embargo; zero index overlap asserted
G9 overfit probability	GREEN	12,870 splits; null calibrates to 0.5 at every S
G10 reproducibility	GREEN	two runs byte-identical: three figures and metrics.json

Invariants — 03 Part B

item	state	note
B1 no network before Gate 0	GREEN	phases 0-4 offline; one deliberate fetch script
B2 every number has an error bar	GREEN	structural in Performance and MetricsReport
B3 append-only trials ledger	GREEN	ledger required on both engines; N never typed
B4 strategies emit weights	GREEN	sizing and execution belong to the engine
B5 both engines identical	GREEN	changed together; G2 re-run in the same commit
B6 no bfill	GREEN	forward-fill only, declared
B7 frozen dataclasses	GREEN	no in-place mutation of Bars or Result
B8 per-observation internally	GREEN	annualised only at the reporting boundary
B9 seeds threaded explicitly	GREEN	no global RNG anywhere; G10 depends on it
B10 non-excess kurtosis	GREEN	fisher=False; asserted against the Lo collapse

Specification documents

item	state	note
00 Gate 0.0 A/B/C	GREEN	exact quadrature, diff 0.00e+00; the compensation figure
00 Gate 0.1-0.5	GREEN	recovery, convergence, signal, degeneracy, determinism
00 phase ordering	GREEN	phases 0 through 7 complete; this is Phase 8+
01 B1 Sharpe and SE	GREEN	Lo (2002) non-normal, plus the Newey-West HAC t
01 B2 PSR	GREEN	deflated.psr
01 B3 deflated Sharpe	GREEN	now fed an N read from the ledger
01 B4 effective trials	GREEN	participation ratio primary, clustering secondary
01 B5 stationary bootstrap	GREEN	twin implementations, bitwise identical
01 B6 PBO via CSCV	GREEN	G9
01 Part C trials ledger	GREEN	implemented; the trap counts 50, not 1
01 Part D reporting contract	GREEN	ten fields, written to metrics.json
01 Part E selection rules	GREEN	four rules; the PBO-vs-temperature figure
02 Parts A-H	GREEN	A4 ruled and verified live; data contract with manifest
03 Parts B, G, H	GREEN	ten invariants held; Part H decisions respected
PLAYBOOK Phase 6 zoo	partial	TSMOM added; MA, mean reversion, vol target, null done
PLAYBOOK Phase 7 cross-sectional	open	N-asset universe and factor attribution
PLAYBOOK Phase 8 reporting	partial	metrics.json done; tearsheet and surfaces open
PLAYBOOK Phase 9 writeup	open	docs/research-note.md

What is next

item	why
Phase 7 — cross-sectional and attribution	N-asset universe, decile long/short, Fama-French plus momentum with Newey-West standard errors. This is the revisit condition Part H decision 1 named for the SPY-only universe.
Phase 8 — tearsheet and parameter surfaces	The in-sample against out-of-sample heatmap PLAYBOOK calls the best figure in the repo.
Phase 9 — research note	docs/research-note.md, structured as a paper, with the limitations section the data biases already require.

Generated 2026-08-24 from 196a5ca. Every statistic in this report is asserted by a test in the tree and carries the standard error it was measured with; none was rounded toward a nicer number.